Leveraging Advanced Analytics to Predict Baseball Production

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Abstract

Within the high-stakes environment of Major League Baseball, characterized by substantial financial investments, the decision-making process of team general managers in player contract signings is very important. This paper presents a comprehensive model aimed at quantifying the peak and decline in production levels of professional baseball players over time. By identifying key variables influencing player performance and synthesizing them into a unified "production" variable, this model offers a systematic approach to understanding and predicting the trajectory of player output. Through statistical analysis and mathematical modeling, we portray the impact of various factors on player performance rise and decline, thereby providing valuable insights for baseball managers in contract negotiations and roster management. The results of this study not only enhance our understanding of player productivity dynamics, but this paper also offers practical implications for optimizing team performance and resource allocation in the highly competitive landscape of professional baseball.

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1 Introduction

Major League Baseball (MLB) is one of the oldest, and most well known, professional sports leagues in the world. The first professional team was founded in 1869, which was named the Cincinnati Red Stockings. Since then, the MLB has undergone many changes in the rules, the field, and even the baseball itself. In the present day there are 30 MLB teams, divided equally between two divisions, the American League (AL) and the National League (NL). The league's rich history is adorned with iconic moments, legendary players, and a cultural significance that extends far beyond the diamond. With its enduring popularity, MLB games are an integral part of American sports culture, attracting and captivating audiences with the timeless allure of America's pastime.

In recent years, MLB has embraced the era of advanced analytics and data-driven decision-making. The league has become a bastion for statistical analysis, employing an extensive array of metrics and technologies to evaluate player performance, make strategic decisions, and gain a deeper understanding of the game. This surge in data utilization has given rise to groundbreaking developments, such as Statcast technology, which tracks player movements and ball trajectories with unparalleled precision. This surge has advanced to the point that Statcast generates around seven terabytes of data for each game. Teams now leverage this data to assess player strengths and weaknesses, optimize lineups, and refine in-game strategies. The marriage of tradition and technology has propelled MLB into an era where data plays an integral role in shaping the narrative of the game, enriching the fan experience, and offering new insights into the sport's intricate dynamics.

2 Data Cleaning

The data selected for this paper is from two separate sources. The first is from Baseball Reference and the second is from Baseball Savant, also known as Statcast. We'll utilize the Baseball Reference dataset to determine the variable we'll designate as our "production" variable, as well as identify which variables correlate most strongly with it. The other dataset, from Baseball Savant, is the data that we will be using to create that production variable, fit our model, as well as predict future player production. We will now go through the steps used to clean the two datasets.

Baseball Reference Cleaning

The Baseball Reference dataset begins with 1560 rows and 54 columns. The first thing that we will do in cleaning this data is remove irrelevant rows or columns. Since the data includes full career data, it also includes two separate "summary" rows of data for that player. The first summary row is a combined total of the full career data. The other summary row is a "162 game average". Every single player in the dataset has these two rows. So, removing them brings the dimensions down to 1390 rows and 54 columns. The next step we took was removing any unnecessary columns. Since this dataset is only going to be used for finding our production variable, we will not need any variables that are not related to the players official stats. Thus we removed these columns: "Pos", "Awards", "Year.additional", "Lg", "EV", "HardH.". The first four variables in this list are character variables that we do not need for the production variable. We will also remove 'EV' and 'HardH.' due to the high number of NA values associated with these official statistics. Our final dimensions of the Baseball Reference dataset is 1390 rows and 48 columns.

The final step in cleaning the Baseball Reference dataset was turning any percentage variables into decimals. To do that, all we will do is take the percentage sign out of the variable, change the variable to a numeric value, then divide the variable by 100. Now that we have cleaned this dataset, we will move on to cleaning our other dataset.

Baseball Savant Cleaning

The Baseball Savant dataset does not need nearly as much cleaning. The only thing that we will do is take out outliers. As we see from figure 2, the plot on the left shows the dataset that has all players included. From this plot we can see that at ages 19, 41, and 42, there are only a few players at each age. The plot on the right shows what the count looks like with those players removed. The player count after removing these outliers is 2590.

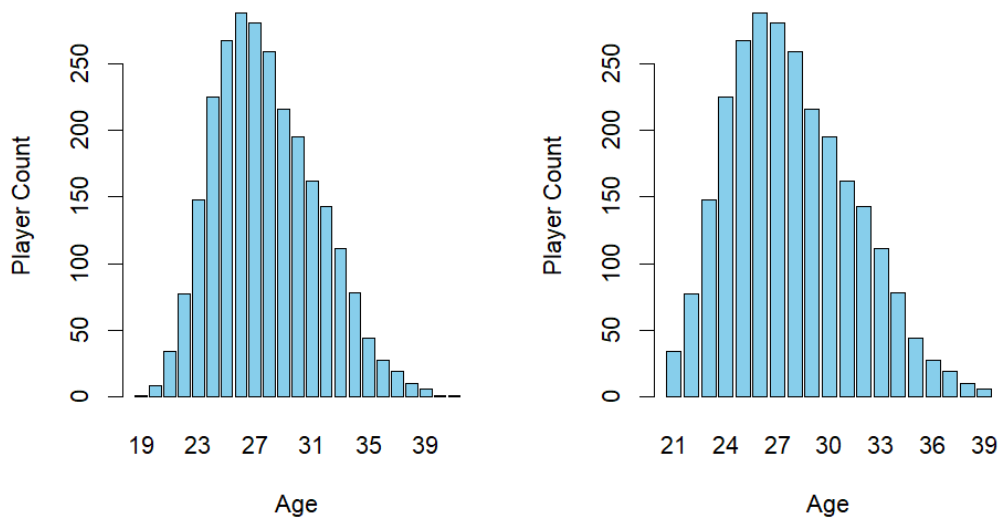

Figure 1: Bar plot of Player Count

Another method for removing outliers is to ensure that players have a minimum number of plate appearances in a season. To do that, we are going to remove any season that a certain player had less than 200 plate appearances. The reason why we are making this adjustment is that when players do not have as many plate appearances, they can skew the data by either having a very hot or very cold streak. When there is a full season of plate appearances, we are able to more accurately predict the way that player will perform in seasons to come. As a result, our player count decreases from 2590 to 2550.

There is one last thing we need to do to this dataset. In the year 2020, each MLB team only played 60 games, instead of the normal 162. So, we will need to take every non-percentage/non-normalized variable and multiply it by 162, then divide it by 60. This normalization process ensures that all of the numeric values in the dataset reflect the players' performance consistently across seasons, compensating for the abbreviated 2020 MLB season. We are assuming that the players will continue to perform at the same rate they did for the last 102 games of the season as they did for the first 60.

3 EDA - Baseball Reference

Looking at the dataset that we have from Baseball Reference, we have full career data from players who have played in the 2002 to 2012 All-Star Games. There are 85 unique players in this dataset. Since they are All-Star players, most of them have had very long careers. Now, looking at our dataset from Baseball Savant, we have all of the players who played in the 2016-2023 seasons, whether they only played one year in that time span, or multiple. There are 796 unique players in this dataset.

The variable that we will be looking at the most, from the Baseball Reference dataset, is named "WPA", also known as "Win Probability Added". We will be using this variable as our "production". However, this variable is not in the Baseball Savant dataset, so, we are going to plot a correlation matrix to identify the variables with the highest correlation to "WPA." We will also create a table showing the variables with the 10 highest correlation values.

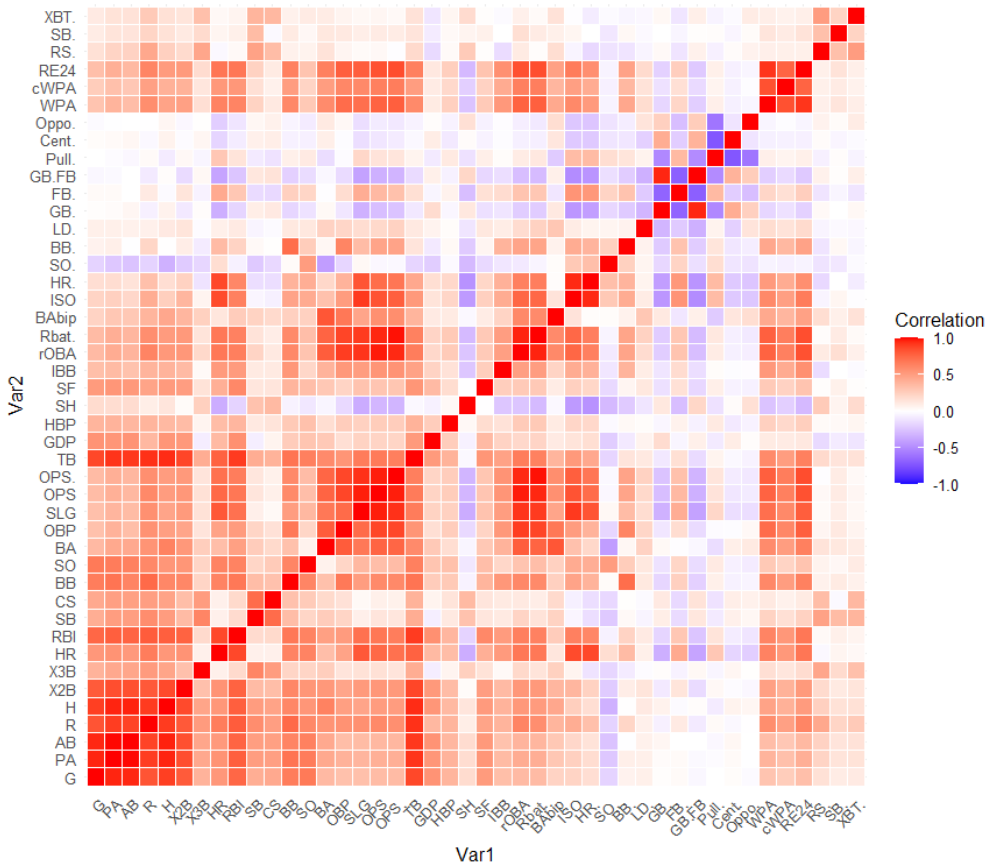

Figure 2: Correlation Plot

As we can see from table 1, there are quite a few variables that are highly correlated with our WPA variable. However, the top correlated variable, RE24, is already a predictive variable that takes WPA into account, so we will not be able to use that.

The first variable we will examine is OPS+, this variable is very highly used in the baseball world as the best hitting variable. OPS+ is a more advanced version of OPS (On base plus slugging). OPS+ takes into account on-base percentage (OBP) and slugging percentage (SLG), and it is then normalized for the league average. To give an example of OPS+, in 2014, Miguel Cabrera had an OPS of .895, so then after adjusting for league and park factors, his OPS+ came out as a 150. That means that in the 2014 season, Miguel Cabrera was 50% better than the average hitter.

Table 1: Correlation Table

Rank	Variable	WPA Correlation
1	WPA	1
2	RE24	.92331463
3	OPS+	.7850286
4	rOBA	.77021358
5	OPS	.76805490
6	OBP	.72949331
7	HR	.63202821
8	ISO	.60010141
9	BA	.5896715
10	HR%	.51790832

The next variable we will look at is ISO (isolated power). This variable is one of the most widely used power stats. This stat will take into account how many doubles, triples, and home runs a player hits. The equation for this stat is very easy: it is the player's slugging percentage (SLG) subtracted by their batting average (BA).

The final variable that we will be looking at is batting average (BA). This is one of the most basic statistics in baseball, however it is still very useful as it is 9th in our correlation table. BA has a very simple equation, the only variables that go into it are the player's hits and at-bats.

These are the three offensive variables that we will be using to make our production variable. This will give us an all-around player combined with the ability to hit for power, the ability to get on base, and the ability to hit for average.

4 EDA - Baseball Savant

Let's now do some analysis on the Baseball Savant dataset. This dataset includes something that the other one does not, and that is defensive variables. To be a complete baseball player, you

need to be good defensively as well, so we will include one of the more used defensive variables into our production variable. This variable is Outs Above Average (OAA). Simply, this statistic gives every player an integer value which shows how much better defensively they are compared to the "average" level. OAA provides valuable insight into a fielder's defensive contributions beyond traditional statistics like fielding percentage or errors. It accounts for factors such as range, reaction time, and positioning, offering a more comprehensive evaluation of defensive prowess.

Moving on, there is one last issue we need to tackle. Our two datasets do not have the exact same variables. In the Baseball Reference dataset, there is a broader range of basic statistics (BA, SLG, AVG), and in Baseball Savant, the data is made up of "expected" data (xBA, xWOBA, xISO). The difference in these stats, is that they get rid of a lot of the luck factor that goes into baseball and just give out the "skill" that a player has. Thus our four variables are giving in table 2 below:

Table 2: Production Variables

Variable Name	**Variable Type**	**Full Name**
OPS	Offensive	On-base% plus Slugging%
xBA	Offensive	Expected Batting Average
xISO	Offensive	Expected Isolated Power
OAA	Defensive	Outs Above Average

The Production Variable

Now that we have looked at all of the variables that we are going to use to make our production variable, it is finally time to make it. The first step to making this variable, is to normalize each of our four variables to ensure that they are all on the same scale. This step involves subtracting the mean of each variable from its values and then dividing by the standard deviation. Normalization standardizes the variables, preventing one variable from dominating the composite score simply because of its scale.

The next step that we will do is assign weights to each variable. Our weighting scheme will

reflect the belief that offensive statistics contribute 60% to a player's overall performance, while defensive metrics contribute 40%. Thus, OAA will be assigned with 40% by itself, as each of the offensive variables with be assigned with 20% each.

The production variable is then calculated by multiplying each normalized variable by its assigned weight and summing the results. This process aggregates the weighted contributions of each variable into a single score, representing the overall player production. The resulting score is then added as a new variable, "CompositeScore", to the dataset for each player, providing a synthesized measure of their performance combining offensive and defensive aspects of their game.

5 Making a Model

Now that we have done EDA on all of our data and our production variable is made, The next step is making our models. There are three different types of regression models that will be created: linear, quadratic, and cubic. To prepare for these models, we will need to find what our predictive variable is and our response variable. The selected predictor is player age, with a focus on understanding its relationship to production. Our response variable will be our production variable.

There is one last thing to do before making our models. We will need to calculate the average of the production variable by age. This will ensure that our model is given one "CompositeScore" value per a player's age. This new variable is created and named "avg_CompositeScore". Figure 5 below shows the average composite score by age.

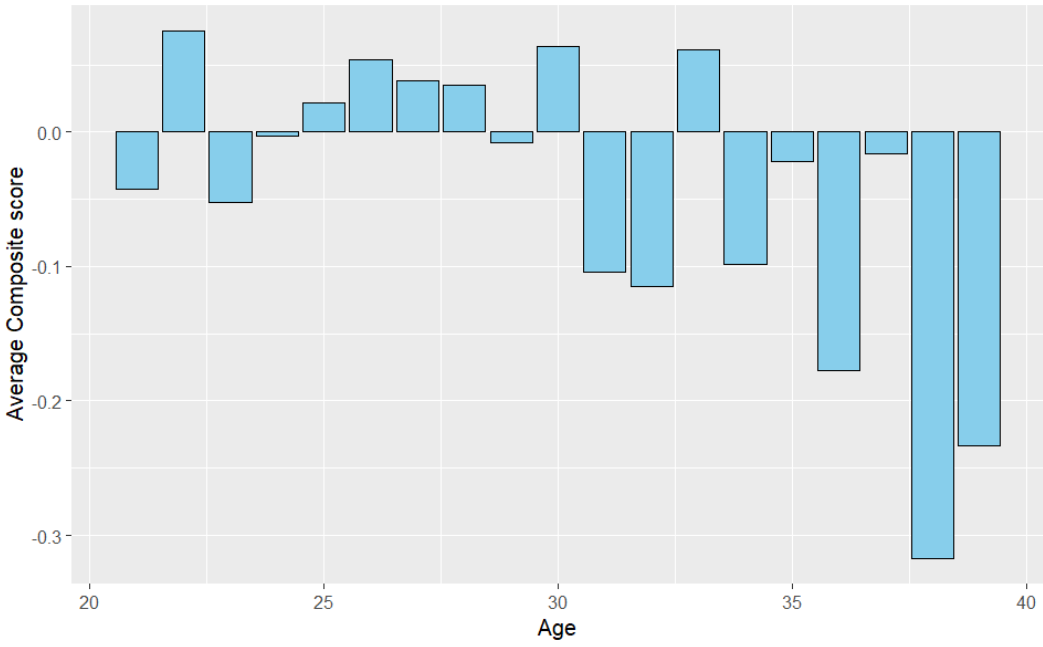

Figure 3: Average Composite Score by Age

Linear Regression Model

For our first model, we will look at a linear regression model. The equation for the linear regression model is shown below. This model is easily explained. As we have said before, avg_CompositeScore is the response variable and player_age is the predictive variable. β_0 will be the value that is given when the player's age is 0, β_1 will be the linear change that happens in the composite score with respect to age. ϵ is the last term that is in this model. The ϵ simply accounts for all factors other than the predictor variables included in the model that affect the dependent variable. These factors could include measurement error, unobserved variables, or random variability in the data that is not accounted for by the model.

$$\text{avg_CompositeScore} = \beta_0 + \beta_1 \times \text{player_age} + \epsilon$$

To create this model, we will use the 'lm' function inside of R. The results of the model creation

are shown below. We now have our equation. β_0 was given the value .3133. β_1 was given the value -.0119. To interpret this, we must go to the most simple terms. The β_1 value tells us that every single year a player plays, the next year he will get a composite score that is .0119 less than his last year.

$$\text{avg_CompositeScore} = .3133 - .0119 \times \text{player_age} + \epsilon$$

The next step that we will do is take a look at this equation over our plot of the average composite score. As you can see in figure 5, our model line does not fit the data as our data does not have a linear trend. When we look at the summary of this model, table 3, there is most of the same story, our R^2 value is 0.3903, which means that 39.03% of the variance is explained by this model. This is not a good value for trying to model the data. We will enhance this model by incorporating additional predictors.

Table 3: Summary of Linear Model

Coefficients	**Estimate**	**Std. Error**	**t-value**	**p-value**
Intercept	0.3133	0.11019	2.843	0.01124 *
player_age	-0.0119	0.00361	-3.299	0.00424 **
Adjusted R^2	0.3544			

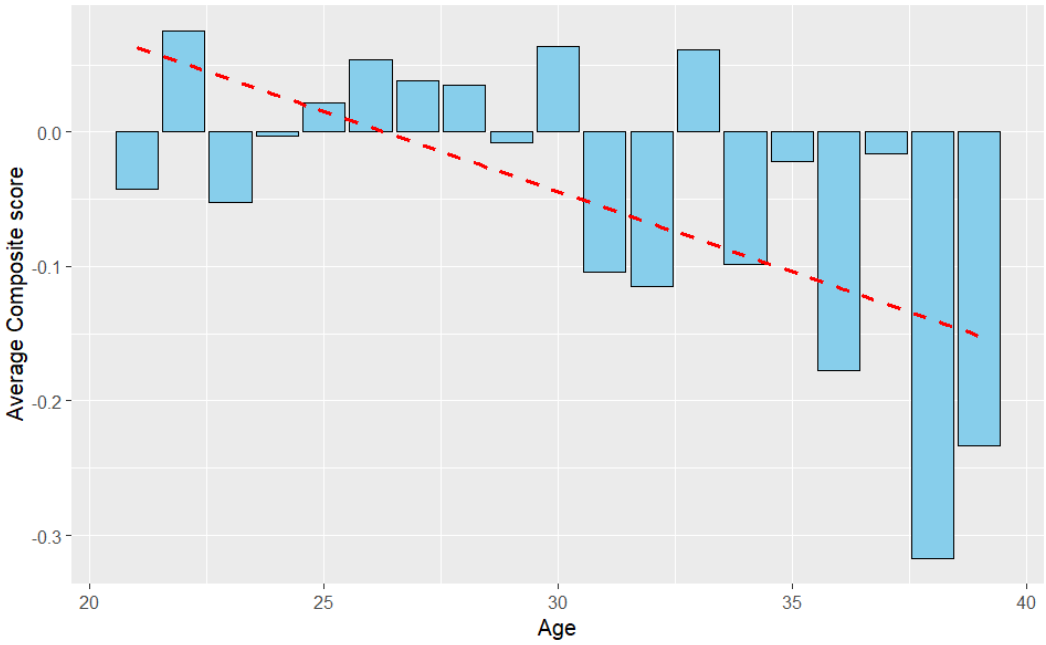

Figure 4: Average Composite Score by Age with Linear Model

Quadratic Regression Model

The next model is a quadratic model. This model will be able to show us a curve over the non-linear relationship between the player's age and their composite score. The equation for this quadratic regression model is shown below:

$$\text{avg_CompositeScore} = \beta_0 + \beta_1 \times \text{player_age} + \beta_2 \times \text{player_age}^2 + \epsilon$$

This model is similar to the linear model as avg_CompositeScore and player_age are still the response and predictive variables. As well as β_0 being the value of the intercept when player_age is 0. β_1 is the linear coefficient and β_2 is our quadratic coefficient. Similar to the linear regression model, the linear coefficient indicates how the average composite score changes linearly with respect to age. A positive β_1 suggests that the average composite score tends to increase as age increases, while a negative β_1 suggests the opposite. The quadratic coefficient indicates how the

rate of change of the average composite score changes as age increases. If β_2 is positive, it suggests that the rate of increase in the average composite score is accelerating with age (concave upward curve), while a negative β_2 indicates deceleration (concave downward curve).

Table 4: Summary of Quadratic Model

Coefficients	**Estimate**	**Std. Error**	**t-value**	**p-value**
Intercept	-.04428	0.01705	-2.596	0.01948 *
player_age	-0.28457	0.07434	-3.828	0.00148 **
player_age^2	-0.19519	0.07434	-2.626	0.01836 *
Adjusted R^2	0.5206			

$$\text{avg_CompositeScore} = -.0443 - 0.2845 \times \text{player_age} - 0.1952 \times \text{player_age}^2 + \epsilon$$

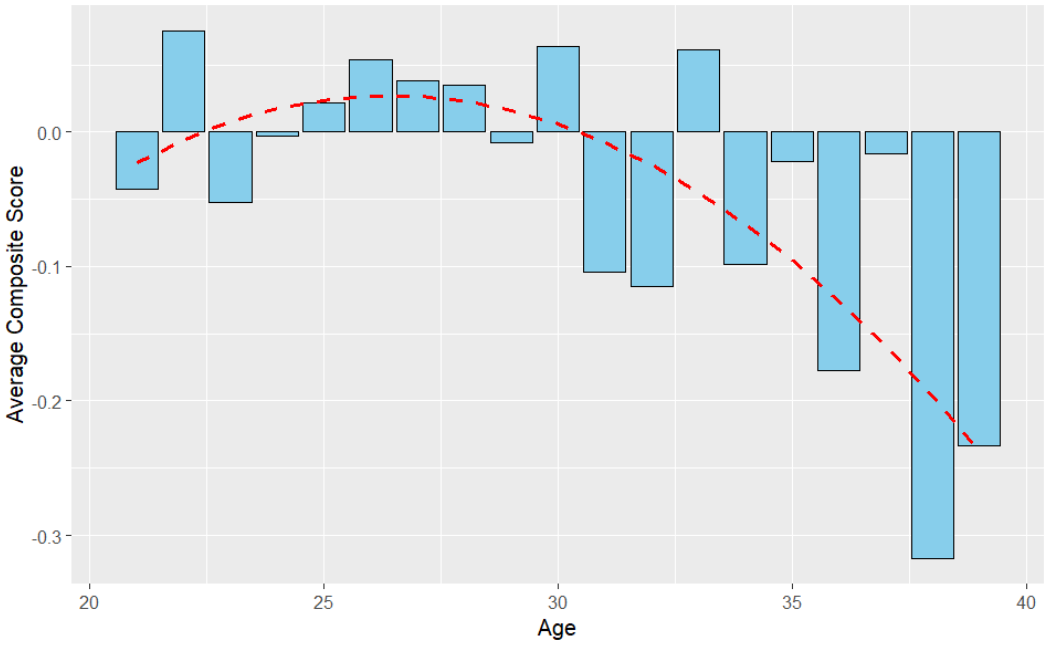

Figure 5: Average Composite Score by Age with Quadratic Model

Looking at figure 5 above, we can see this model fits the data much better. This model catches the slight upwards motion at the beginning as well as the downfall of the composite score. We

can confirm this by looking at the summary in table 4. The R^2 value that we get from this model is also much improved, standing at 0.5739. This is telling us that 57.39% of the variance is explained by the model. However, one of the most important columns to look at in this table is the last column. This column shows the p-value given to each coefficient. As we can see each of the p-values are under .02 which makes the statistically significant. R also gives us a nice indicator to tell if a coefficient is statistically significant. As we can see, after the "p-value" column, there are asterisks. These will tell us if a coefficient is significant. The rule for these asterisks, is that the more asterisks there are, the more statistically significant the predictor is.

Cubic Regression Model

The last model we will look at is a cubic regression model. We are looking to see if this model will add any valuable insight to our data. The equation for the cubic regression model is shown below:

$$\text{avg_CompositeScore} = \beta_0 + \beta_1 \times \text{player_age} - \beta_2 \times \text{player_age}^2 - \beta_3 \times \text{player_age}^3 + \epsilon$$

For this model, β_1 and β_2 act the same as the model before. However, we have added a new coefficient. β_3 is our cubic coefficient. The cubic coefficient indicates how the rate of change of the average composite score changes as age increases or decreases. A positive β_3 suggests that the rate of increase or decrease in the average composite score is accelerating or concave upward, respectively, with age. Conversely, a negative β_3 indicates deceleration or concave downward behavior.

$$\text{avg_CompositeScore} = -.0443 - .2846 \times \text{player_age} + .1952 \times \text{player_age}^2 + .0269 \times \text{player_age}^3 + \epsilon$$

Table 5: Summary of Cubic Model

Coefficients	**Estimate**	**Std. Error**	**t value**	**p value**
Intercept	-.04428	0.01754	-2.596	0.02335 *
player_age	-0.28457	0.07646	-3.722	0.00205 **
player_age^2	-0.19519	0.07646	-2.553	0.02208 *
player_age^3	-0.02694	0.07646	-0.352	0.72954
Adjusted R^2	0.4928			

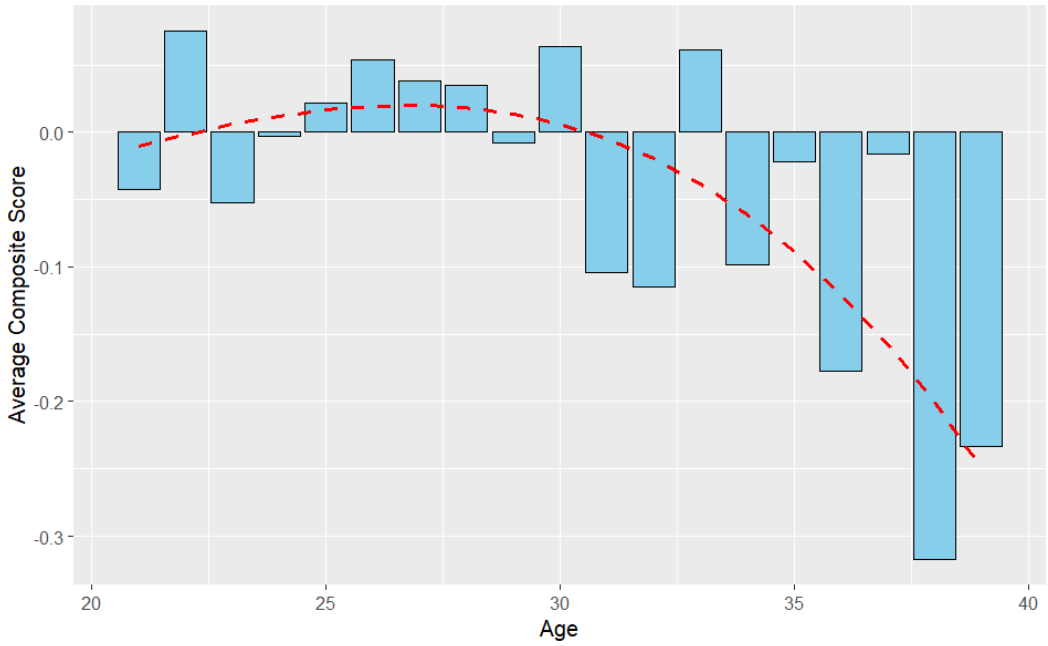

Figure 6: Average Composite Score by Age with Cubic Model

When we look at the cubic model's summary in table 5, we will see a lot of the same as the quadratic model. One thing that we need to notice, is that the last coefficient in the table is given a .72954 p-value. A high p-value for the cubic coefficient indicates the cubic term may not have a significant impact on the response variable after accounting for the linear and quadratic terms. This suggests that we should go with our quadratic model.

Normalized Data

Now that we have found the model that works best with the "CompositeScore" variable, there is one more way that we can make this model better. To do this we will do a calculation on every player's unique composite scores. The calculation that we will be doing is called "mean normalization". To do this calculation, we will first compute the average composite score for each unique player across all years. We will then subtract the composite score for each year from the average score. Figure 5 shows the plot of the new values.

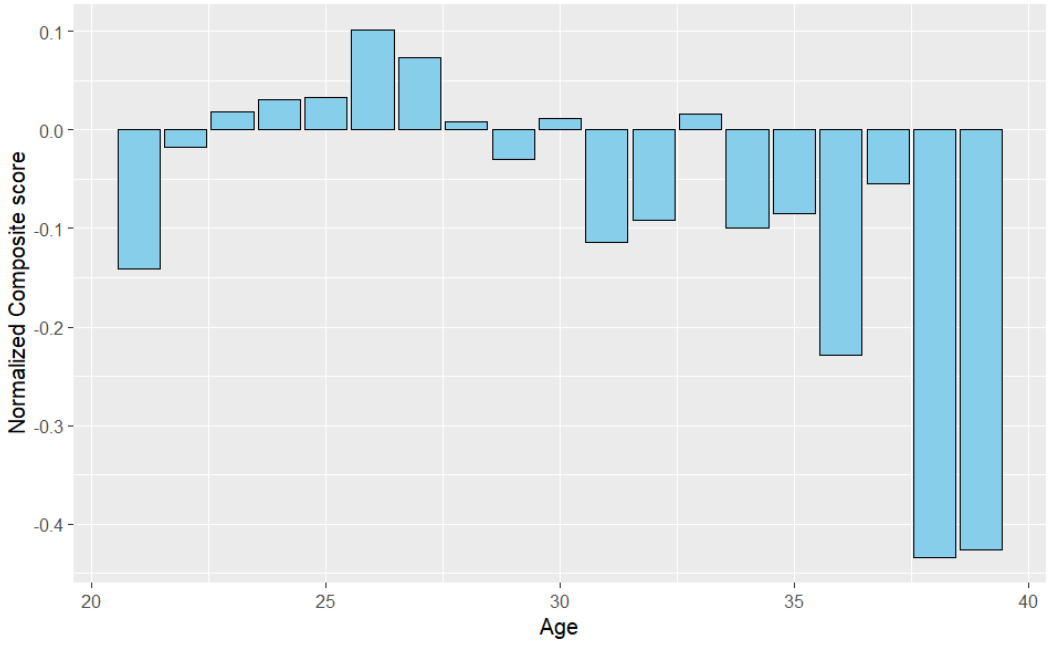

Figure 7: Normalized Composite Score by Age

When we remove bias from player output, we aim to isolate the raw skill or talent of the player from other factors that might affect performance, such as external conditions, randomness, or luck. In the context of baseball, where individual player performance can be influenced by various factors beyond their control, removing bias can help us assess the true skill level of a player.

By centering the data around the player's average performance, we are essentially looking at

how a player's performance differs from their typical or expected level. This can help us identify trends, patterns, or improvements that are more likely attributable to the player's skill rather than random fluctuations or external factors.

$$\text{avg_CompositeScore} = -0.0751 - 0.4101 \times \text{normalized_player_age} - 0.3586 \times \text{player_age}^2 + \epsilon$$

Table 6: Summary of Normalized Quadratic Model

Coefficients	**Estimate**	**Std. Error**	**t-value**	**p-value**
Intercept	-0.07508	0.01795	-4.183	0.000703 ***
player_age	-0.41014	0.07823	-5.243	8.05e-05 ***
player_age^2	-0.35855	0.07823	-4.583	0.000306 ***
Adjusted R^2	0.7209			

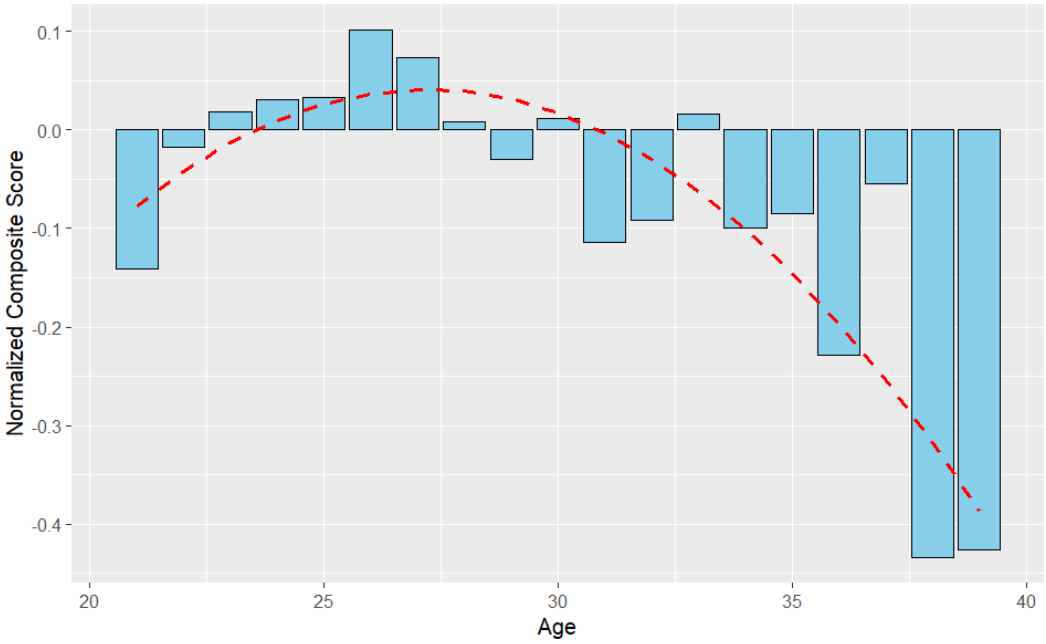

Figure 8: Normalized Composite Score by Age with Quadratic Model

As we can see from table 6 and figure 5, our model fits the data much better than the other models. The p-values of our coefficients all have three asterisks, which mean that all of our

predictors are highly significant to predicting our composite score. The R^2 of our model is also much higher than the other models, sitting at .7209.

One of the more surprising aspects of this model is the value for β_2. This, as we learned before, is the value at which the rate of increase or decrease over time. After removing the bias in the model and changing to raw skill, this value decreased from a '-0.19519' to '-0.35855'. Suggesting that there is a more rapid decline in skill than what our non-normalized data is showing.

What we want to learn from this, is where is there the most risk to sign certain players during their career. These models give us a very good look into where the player's peak performance is, and where there decline starts. Every single model that we created has peaked in the 26-28 age range. Our last model is showing that they are also playing at a high level into their age 31 season. There is one more way we will test this hypothesis, and that is with a T-test.

6 T-Test

A t-test is a statistical hypothesis test used to determine whether there is a significant difference between the means of two groups. In our case, we will use it to find out at what age difference there is a significant statistical difference in the average composite score.

First, we will go over our two hypotheses. Our null hypothesis is that there is no significant difference between ages. This means that all of the difference that we see in the graph is all due to random variation or chance. Our alternative hypothesis is that there is significant difference between the ages. This implies that the observed difference in composite score is not due to random chance, but due to the players aging, or regressing in skill.

To be more specific, we will be doing a δ_3 t-test. This means that we will be looking at the change of 3 units of time. In our case, we will be looking at a change of 3 years of age. So, if our first age is 21 years old, the second age will be 24 years old. To do this, we will run a function in R which takes in our dataset, the younger age, the older age, and then a variable to do the t-test on. When we run this, we will use the "CompositeScore" variable.

There are three steps that we will go through for this test. The first step is identifying all unique players who had a season with both ages. For example, if we are running a test on age 22, we will only run the test with players who, in our dataset, has season where they were 22 and 25 years old. The next step is finding the difference between their "CompositeScore" value for those seasons. The last step is running the "t.test()" function in R. We will run this test over all ages. Figures 6, 6 and table 7 show the unique T-Values and P-Values for every unique age grouping.

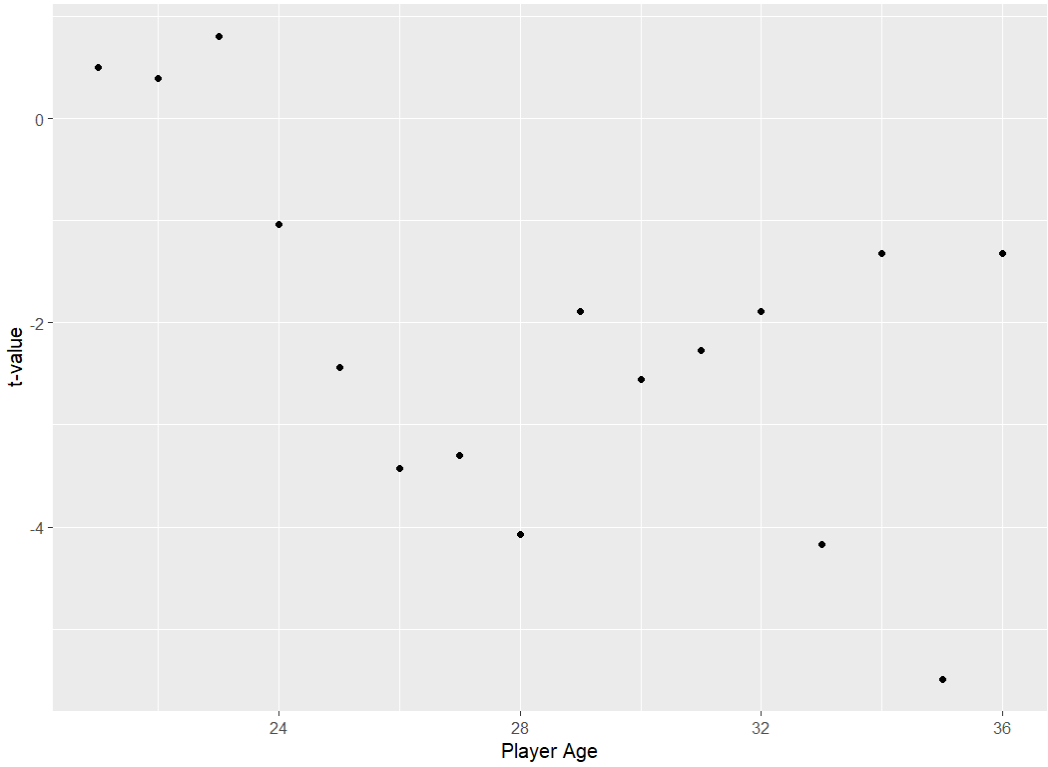

Figure 9: T-Values from T-Test

Table 7: T-Values

Age Pair	**T-value**	**P-value**
21 - 24	0.4985818	0.6253054965
22 - 25	0.3896567	0.6993717906
23 - 26	0.8084335	0.4222622497
24 - 27	-1.0360073	0.3036657996
25 - 28	-2.4370223	0.0166907004
26 - 29	-3.4220985	0.0009289693
27 - 30	-3.2988559	0.0014312766
28 - 31	-4.0725006	0.0001154438
29 - 32	-1.8885574	0.0641347072
30 - 33	-2.5586462	0.0137173423
31 - 34	-2.2718658	0.0299554920
32 - 35	-1.8906083	0.0732546267
33 - 36	-4.1693638	0.0009452356
34 - 37	-1.3191287	0.2236303520
35 - 38	-5.4958879	0.0118542326
36 - 39	-1.3246452	0.4116650428

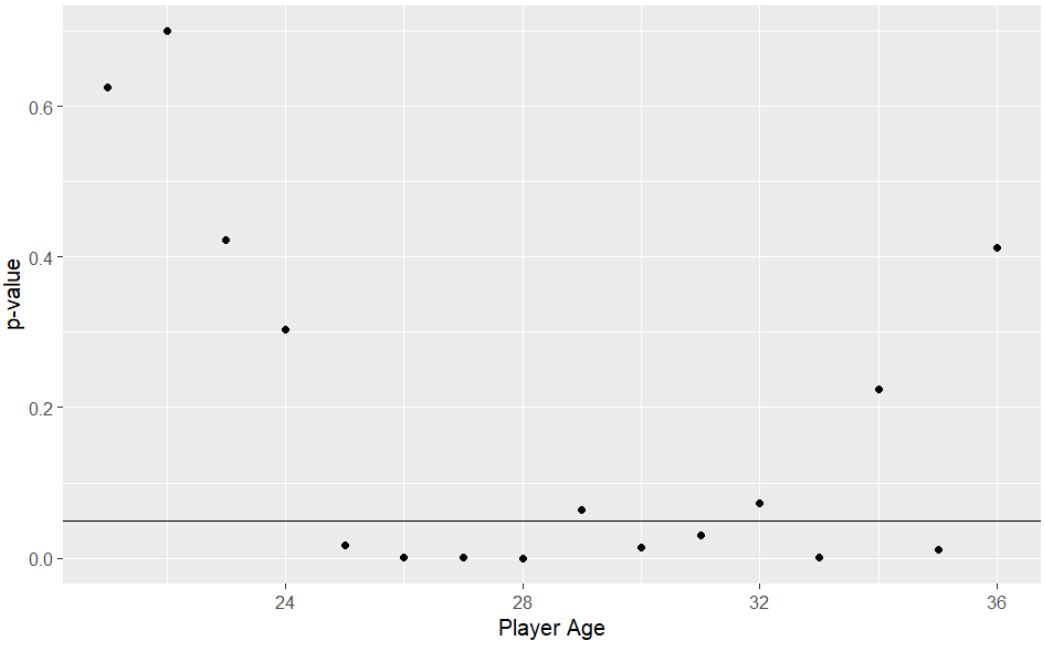

Figure 10: P-Values from T-Test

To explain these plots, we must first know what we are looking at. The t-value is used to assess the statistical significance of the coefficient estimate. A large t-value indicates that the coefficient estimate is unlikely to be zero, suggesting that the predictor variable, player age, has a significant effect on the response variable, composite score. Conversely, a small t-value suggests that the coefficient estimate is not significantly different from zero, indicating that the predictor variable may not be important in explaining the variation in the response variable. In our plot of t-values, there are negative values. This means that the relationship with composite score and player age is negative.

For the younger age groups, there is not much statistical significance in the changes between years, as we can see from table 7. Age pairs 21-24 all the way to 24-27 have a high p-value, which indicates that either there is not enough data in this area, or the data is non-conclusive if there is a certain trend there. But, when we move to the next few age pairs, from 25-28 all of the way to 33-36, the p-values are almost all under .05. There is a solid line in figure 6 to show this benchmark. This indicates that this section of the data is statistically significant in showing the trend of the composite score. Figure 6 shows that the first large t-value is at the age 25-28 pair.

This t-test is also pointing towards the peak of the baseball players output is around the 26-28 years old range.

7 Conclusion

In conclusion, our exploration into Major League Baseball (MLB) player statistics has provided valuable insights into the game. Through data analysis, we dove into the intricate relationship between various statistical metrics and player performance. Our comprehensive tests revealed compelling trends and patterns that shed light on the dynamics of professional baseball. One of the key discoveries of our study is the significant influence of age on player performance. By analyzing the data, we observed that players typically reach their peak performance levels in their late twenties, typically around 26 to 28 years old. This finding underscores the importance of age as a crucial factor in understanding player capabilities and longevity in the MLB.

Understanding the age at which players typically peak can profoundly impact how managers structure player contracts. Armed with this knowledge, managers can negotiate contracts that align with a player's prime years, ensuring optimal performance and value for the team. For instance, rather than offering long-term contracts to players who have already passed their peak, teams may opt for shorter, more performance-based contracts. Conversely, for players approaching their peak or in the midst of it, teams may be more inclined to invest in longer-term deals to secure their services during their prime playing years.

In essence, our research highlights the transformative power of data analytics in modern baseball. By leveraging statistical insights, teams can optimize their performance and gain a competitive edge in the ever-evolving landscape of professional sports. As we continue to harness the power of data-driven methodologies, the future of baseball promises new horizons of innovation and excellence.

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